

Introduction

How do people understand bar graphs?

- Ceja et al. (2021) had participants view vertical bars of equal length presented in three widths: wide, square, narrow.
- They overestimated the length of wide bars and underestimated that of narrow bars.
- These findings suggest that people process the length *and* width of bars to understand the numerical information conveyed in bar graphs.
- More generally, these dimensions are *magnitudes*.
- Thus, people understand the magnitudes depicted in the bar graphs.

How, then, do people understand the magnitudes?

- This question is central to the field of numerical cognition.
- A critical findings is that people represent magnitudes as points on a **mental number line**:

- a horizontally oriented continuum
- a left-to-right ordering of numbers.



Figure 1. An example of number line.

What, then, is the evidence for a mental number line?

- Evidence comes from studies using the comparison task, where participants see two symbolic numbers on a screen and judge which one is greater.
- The **order effect** is that people make faster judgement when the smaller number was presented on the left side of the screen and the larger number on the right side:



Figure 2. Examples of number comparison task and order effect.

Research Questions

The current study investigates if people recruit magnitude representations to understand the numerical values depicted in bar graphs.

RQ1: Is it easier to compare bars that are oriented horizontally (i.e., parallel to the x-axis), because this aligns with the mental number line representation, than bars that are oriented vertically (i.e., parallel to the y-axis)?

RQ2: For vertically oriented bar graphs, is it easier to judge which bar is greater when they are in increasing order left-to-right? Is there no such effect for horizontally oriented bar graphs because there is no preference for increasing top-to-bottom (or bottom-to-top) order?

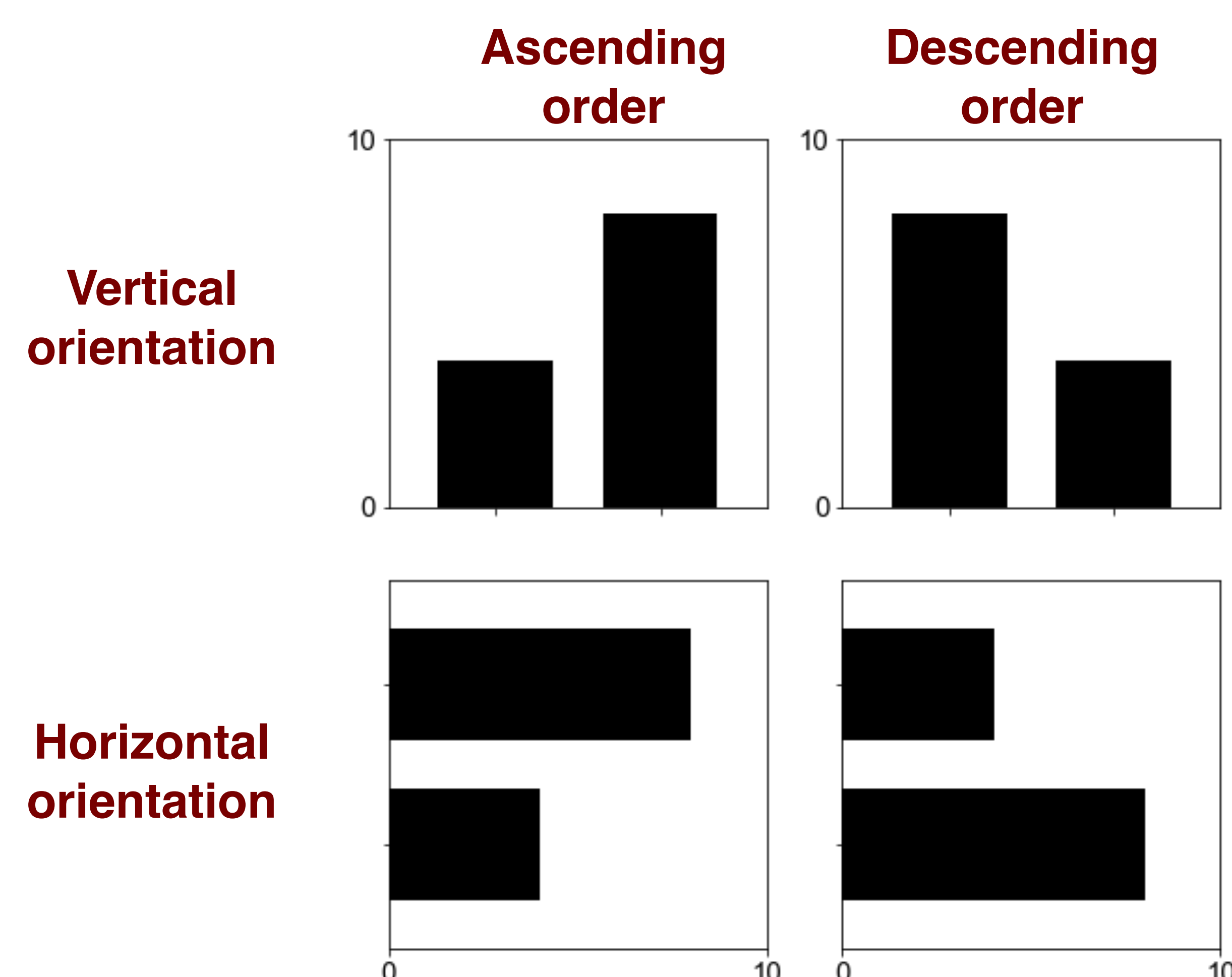
Methods

Participants

We will recruit 40 undergraduate participants in Fall 2022.

Design and Procedure

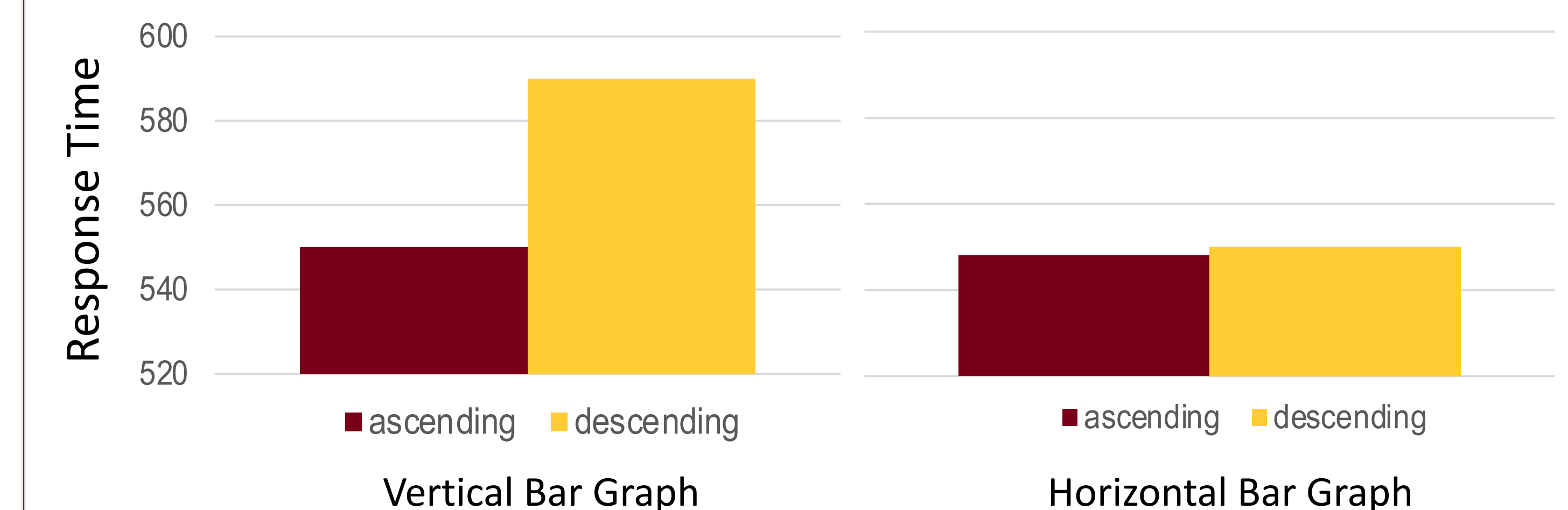
Participants will be asked to judge which of the two bars in a bar graph is greater. There will be block of vertically oriented graphs and a block of horizontally oriented graphs. Within the vertical bar graphs, the order of bars will be presented in either ascending left-to-right order or descending right-to-left order. Within the horizontal bar graph, the order of bars will be presented in either ascending bottom-to-top order or descending top-to-bottom order.



Expected Results

Prediction 1: People will make faster and more accurate comparisons for horizontally oriented bar graphs, consistent with the horizontal mental number line. The alternative hypothesis is also possible: People will perform better for vertically oriented bar graphs due to their ubiquity.

Prediction 2: There will be a bar orientation x bar order interaction. Specifically, people will show the order effect for vertically oriented bar graphs, when the ascending order of the bars matches their positions on the mental number line. By contrast, there will be no order effect for horizontally oriented bar graphs.

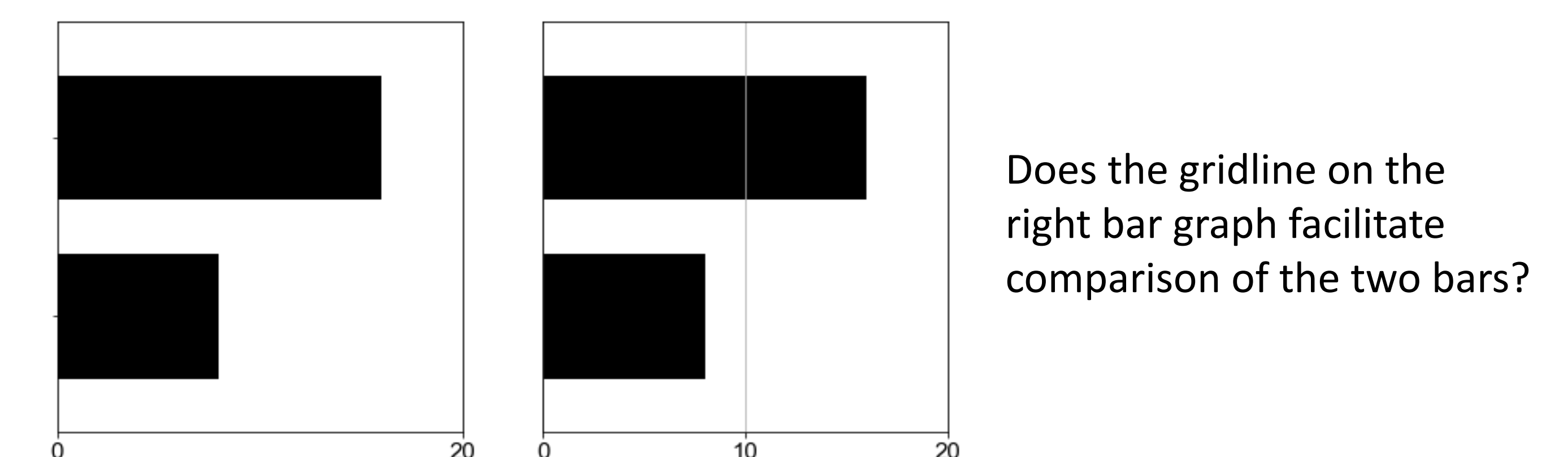


Discussion

If the current study finds that people recruit magnitude representations to understand numerical quantities in bar graphs, we will examine other numerical cognition effects.

Boundary Effect

People have natural boundaries on their mental number line, e.g., 0, 50, and so on. We investigate if gridline on bar graphs serve as natural boundaries and facilitate comparison.



References

- [1] C. R. Ceja, C. M. McColeman, C. Xiong, and S. L. Franconeri, "Truth or square: Aspect ratio biases recall of position encodings," *IEEE Trans. Vis. Comput. Graph.*, vol. 27, no. 2, pp. 1054–1062, 2021.
- [2] E. Turconi, B. Jemel, B. Rossion, and X. Seron, "Electrophysiological evidence for differential processing of numerical quantity and order in humans," *Brain Res. Cogn. Brain Res.*, vol. 21, no. 1, pp. 22–38, 2004.